

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

[2]: df = pd.read_csv(r"C:\Users\Adein\online_learning_platform.csv")
```

1: Dataset Understanding

Display first 5 records

[3]:	df.head()									
[7]:	user_id	age_group	role	primary_platform	learning_device	hours_per_week	learning_purpose	content_type_preference	completion_rate_percent	sat
	0	1	35-44	Working Professional	edX	Laptop	10	Career Growth	Text	34
	1	2	45+	Other	edX	Mobile	12	Career Growth	Text	17
	2	3	25-34	Freelancer	Skillshare	Desktop	2	Academic	Text	67
	3	4	45+	Working Professional	Udemy	Mobile	3	Career Growth	Mixed	25
	4	5	45+	Working Professional	Khan Academy	Mobile	10	Hobby	Video	81

Find number of rows and columns

```
[4]: df.shape

[6]: (428, 10)
```

Display column names

```
[5]: df.columns

[10]: Index(['user_id', 'age_group', 'role', 'primary_platform', 'learning_device',
        'hours_per_week', 'learning_purpose', 'content_type_preference',
        'completion_rate_percent', 'satisfaction_score'],
        dtype='str')
```

Check data types

```
[6]: df.dtypes

[8]: user_id          int64
age_group         str
role              str
primary_platform  str
learning_device   str
hours_per_week    int64
learning_purpose    str
content_type_preference  str
completion_rate_percent  int64
satisfaction_score  int64
dtype: object
```

Complete dataset information

```
[7]: df.info()

<class 'pandas.DataFrame'>
RangeIndex: 428 entries, 0 to 427
Data columns (total 10 columns):
 #   Column              Non-Null Count  Dtype
--  --
 0   user_id             428 non-null    int64
 1   age_group           428 non-null    str
 2   role                428 non-null    str
 3   primary_platform    428 non-null    str
 4   learning_device     428 non-null    str
 5   hours_per_week      428 non-null    int64
 6   learning_purpose      428 non-null    str
 7   content_type_preference  428 non-null    str
 8   completion_rate_percent  428 non-null    int64
 9   satisfaction_score   428 non-null    int64
dtypes: int64(4), str(6)
memory usage: 39.9 KB
```

The dataset contains 420 users and 10 variables. It includes information about user demographics, platform usage, learning device, weekly learning hours, learning purpose, preferred content type, course completion rate and satisfaction score.

2: Data Preparation

Check missing values

```
[8]: df.isnull().sum()

[8]: user_id          0
age_group          0
role              0
primary_platform  0
learning_device   0
hours_per_week    0
learning_purpose    0
content_type_preference  0
completion_rate_percent  0
satisfaction_score  0
dtype: int64
```

Check duplicate records

```
[9]: df.duplicated().sum()

[9]: np.int64(0)
```

Check data types

```
[10]: df.dtypes

[10]: user_id          int64
age_group         str
role              str
primary_platform  str
learning_device   str
hours_per_week    int64
learning_purpose    str
content_type_preference  str
completion_rate_percent  int64
satisfaction_score  int64
dtype: object
```

Check unique values

```
[11]: print(df["age_group"].unique())
print(df["role"].unique())
print(df["primary_platform"].unique())
print(df["learning_device"].unique())
print(df["learning_purpose"].unique())
print(df["content_type_preference"].unique())

<ArrayStringArray>
['35-44', '45+', '25-34', '18-24', 'Under 18']
Length: 5, dtype: str
<ArrayStringArray>
['Working Professional', 'Other', 'Freelancer', 'Student']
Length: 4, dtype: str
<ArrayStringArray>
['edX', 'Skillshare', 'Udemy', 'Khan Academy', 'YouTube', 'Coursera']
Length: 6, dtype: str
<ArrayStringArray>
['Laptop', 'Mobile', 'Desktop', 'Tablet']
Length: 4, dtype: str
<ArrayStringArray>
['Career Growth', 'Academic', 'Hobby', 'Skill Upgrade']
Length: 4, dtype: str
<ArrayStringArray>
['Text', 'Mixed', 'Video']
Length: 3, dtype: str
```

Check numerical ranges

```
[12]: print(df["hours_per_week"].min())
print(df["hours_per_week"].max())

print(df["completion_rate_percent"].min())
print(df["completion_rate_percent"].max())

print(df["satisfaction_score"].min())
print(df["satisfaction_score"].max())

1
20
10
100
1
10
```

I checked the dataset for missing values, duplicate records, incorrect data types and unusual values. No missing values or duplicate records were found. The numerical variables were already stored in suitable formats, so no major data cleaning was required.

Exploratory Data Analysis

1. Distribution of users across age groups

```
[13]: df["age_group"].value_counts()

[13]: age_group
35-44      98
Under 18   91
45+       77
25-34     77
18-24     77
Name: count, dtype: int64
```

The 35-44 age group has the highest number of users with 98 users. The distribution shows that the platform is being used by people from different age groups rather than being concentrated in only one age segment.

2. Hours Spent on Learning Per Week

```
[14]: df["hours_per_week"].describe()

[14]: count      428.000000
mean       10.476190
std        5.700253
min         1.000000
25%        5.750000
50%       11.000000
75%       15.000000
max       20.000000
Name: hours_per_week, dtype: float64

[15]: df["hours_per_week"].mean()

[15]: np.float64(10.476190476190476)

[16]: print("Minimum:", df["hours_per_week"].min())
print("Maximum:", df["hours_per_week"].max())

Minimum: 1
Maximum: 20
```

Users spend an average of approximately 10.48 hours per week on learning activities. The learning time varies between users, showing that some users spend considerably more time on the platform than others.

3.Platform Usage Patterns

```
[17]: df["primary_platform"].value_counts()

[17]: primary_platform
Skillshare    79
Khan Academy  75
Coursera      69
Udemy         68
edX           66
YouTube       63
Name: count, dtype: int64

[18]: df.groupby("primary_platform")["hours_per_week"].mean().sort_values(
        ascending=False)

[18]: primary_platform
edX      11.243424
Udemy    11.182941
Coursera 10.565217
Skillshare 10.341772
Khan Academy  9.986667
YouTube      9.746892
Name: hours_per_week, dtype: float64
```

Users are distributed across multiple online learning platforms. The average learning time also differs between platforms, which suggests that users may interact with different platforms at different levels of intensity.

4.Preferred Learning Content Types

```
[19]: df["content_type_preference"].value_counts()

[19]: content_type_preference
Text      148
Video    146
Mixed     126
Name: count, dtype: int64
```

Users show different preferences for learning content such as text, video and other formats.

Engagement Analysis

1. Average learning hours by age group

```
[20]: df.groupby("age_group")["hours_per_week"].mean().sort_values(
        ascending=False)

[20]: age_group
25-34      11.623377
45+       10.483312
18-24     10.272727
Under 18   10.498981
35-44     10.131367
Name: hours_per_week, dtype: float64

25-34 is having 11.62 hours/week
```

So this age group has the highest average learning time.

2. Relationship Between Learning Hours and Completion Rate

```
[21]: correlation = df["hours_per_week"].corr(
        df["completion_rate_percent"])

print("Correlation:", correlation)

Correlation: -0.83932379948612382
```

A correlation close to 0 means there is almost no linear relationship between the two variables.

So, spending more hours on the platform does not automatically mean that a user will have a higher completion rate.

3. Comparison of platform engagement across different platforms or devices

```
[22]: # Engagement Across Different Platforms
platform_engagement = df.groupby(
    "primary_platform"
)["hours_per_week"].mean().sort_values(ascending=False)

print(platform_engagement)

primary_platform
edX      11.243424
Udemy    11.182941
Coursera 10.565217
Skillshare 10.341772
Khan Academy  9.986667
YouTube      9.746892
Name: hours_per_week, dtype: float64
```

Engagement levels vary across learning platforms. Users of some platforms spend more time learning each week, which may indicate differences in course structure, content quality, user motivation or platform experience.

```
[23]: #Engagement Across Different Devices
device_engagement = df.groupby(
    "learning_device"
)["hours_per_week"].mean().sort_values(ascending=False)

print(device_engagement)

learning_device
Laptop    11.072397
Tablet    10.333333
Desktop   10.109890
Mobile    9.705766
Name: hours_per_week, dtype: float64
```

Laptop users appear to be more engaged when engagement is measured by weekly learning hours. This could indicate that users prefer larger screens for longer learning sessions.

Course Completion Analysis

1. Overall Course Completion Rate

```
[24]: df["completion_rate_percent"].mean()

[24]: np.float64(53.93571428571428)

[25]: df["completion_rate_percent"].describe()

[25]: count      428.000000
mean       53.935714
std       26.630357
min        10.000000
25%       30.000000
50%       54.000000
75%       77.250000
max       100.000000
Name: completion_rate_percent, dtype: float64
```

The average course completion rate is approximately 53.94%. This means that on average users complete around half of their assigned or enrolled course content.

2. Relationship Between Engagement and Completion

```
[26]: df["engagement_level"] = pd.cut(
        df["hours_per_week"],
        q=3,
        labels=["Low", "Medium", "High"])

[27]: engagement_completion = df.groupby(
        "engagement_level",
        observed=True
    )["completion_rate_percent"].mean()

print(engagement_completion)

engagement_level
Low      57.264882
Medium   56.872848
High     54.778659
Name: completion_rate_percent, dtype: float64
```

3. Impact of Learning Purpose

```
[28]: completion_purpose = df.groupby(
        "learning_purpose"
    )["completion_rate_percent"].mean().sort_values(
        ascending=False)

print(completion_purpose)

learning_purpose
Skill Upgrade    59.289524
Academic         53.782178
Career Growth    52.681034
Hobby            49.520721
Name: completion_rate_percent, dtype: float64

Skill Upgrade is approximately 59.21%
```

This is the highest average completion rate among the learning purpose groups.

Users learning for skill improvement have the highest average completion rate, at around 59.21%. This suggests that users with a specific and practical learning goal may be more likely to continue and complete their courses.

User Satisfaction Analysis

1. Relationship Between Completion Rate and Satisfaction

```
[29]: correlation = df["completion_rate_percent"].corr(
        df["satisfaction_score"])

print("Correlation:", correlation)

Correlation: 0.8225716826488458
```

Users with higher completion rates are not necessarily more satisfied than users with lower completion rates.

2. Satisfaction by Age Group

```
[31]: satisfaction_age = df.groupby(
        "age_group"
    )["satisfaction_score"].mean().sort_values(
        ascending=False)

print(satisfaction_age)

age_group
25-34      5.662338
35-44      5.295918
45+        5.285714
Under 18   5.498981
18-24      4.885195
Name: satisfaction_score, dtype: float64
```

Satisfaction varies across age groups. The 25-34 age group has the highest average satisfaction score at approximately 5.66 out of 10.

3.Satisfaction by Platform

```
[5]: satisfaction_platform = df.groupby(
        "primary_platform"
    )["satisfaction_score"].mean().sort_values(
        ascending=False)

print(satisfaction_platform)

primary_platform
Khan Academy    5.693333
Skillshare      5.448354
edX             5.237576
Coursera        5.144928
YouTube         5.126884
Udemy           5.102235
Name: satisfaction_score, dtype: float64
```

Satisfaction differs across learning platforms. Khan Academy users have the highest average satisfaction score at approximately 5.69 out of 10. This difference could be related to content, usability or the type of learning experience offered, but the dataset alone cannot determine the exact cause.

Data Visualization

Engagement Level Distribution

```
[6]: df["engagement_level"] = pd.cut(
        df["hours_per_week"],
        q=3,
        labels=["Low", "Medium", "High"])

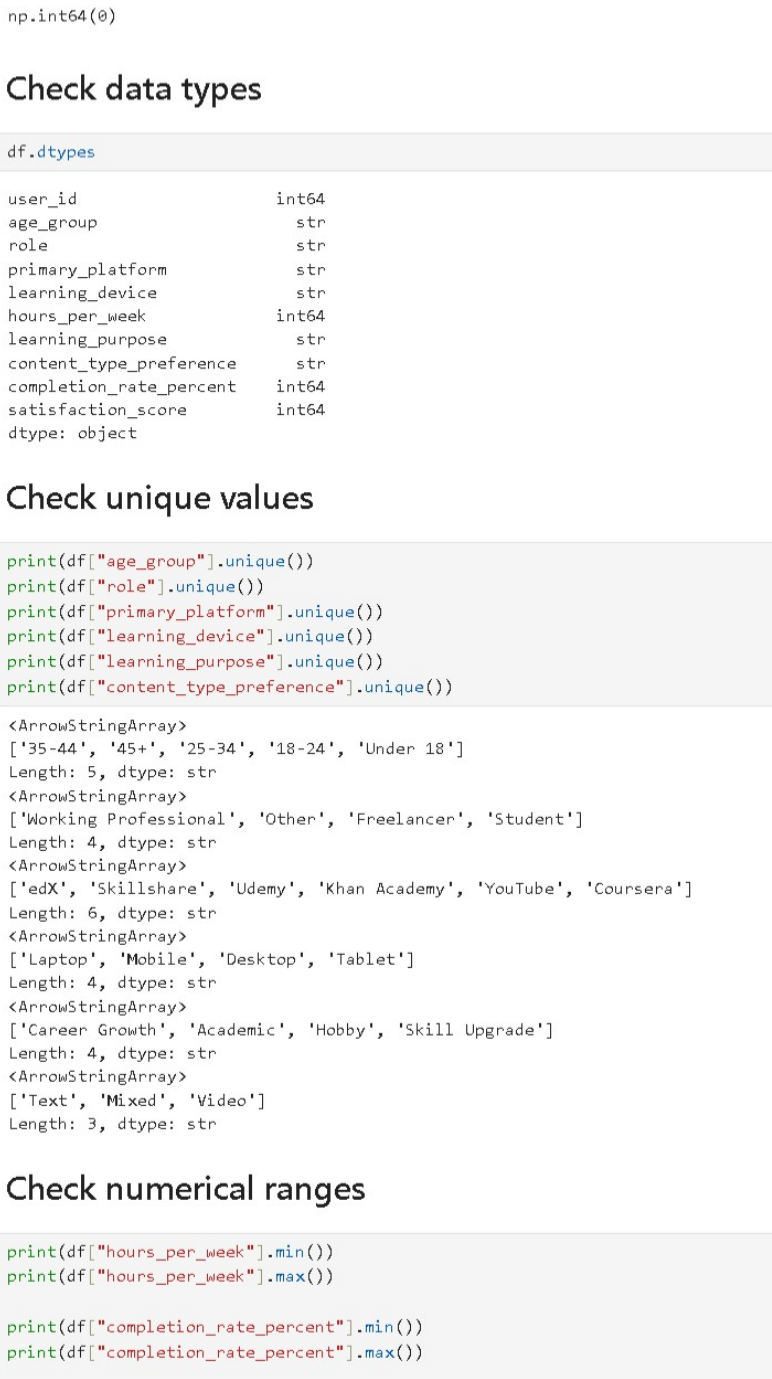
[7]: df["engagement_level"].value_counts()

[7]: engagement_level
Medium    151
Low       147
High      122
Name: count, dtype: int64

[8]: df["engagement_level"].value_counts().sort_index().plot(
        kind="bar")

plt.title("Distribution of Users by Engagement Level")
plt.xlabel("Engagement Level")
plt.ylabel("Number of Users")
plt.xticks(rotation=0)

plt.show()
```



This chart divides users into low, medium and high engagement groups based on their weekly learning hours. It helps identify how the overall user base is distributed according to learning activity.

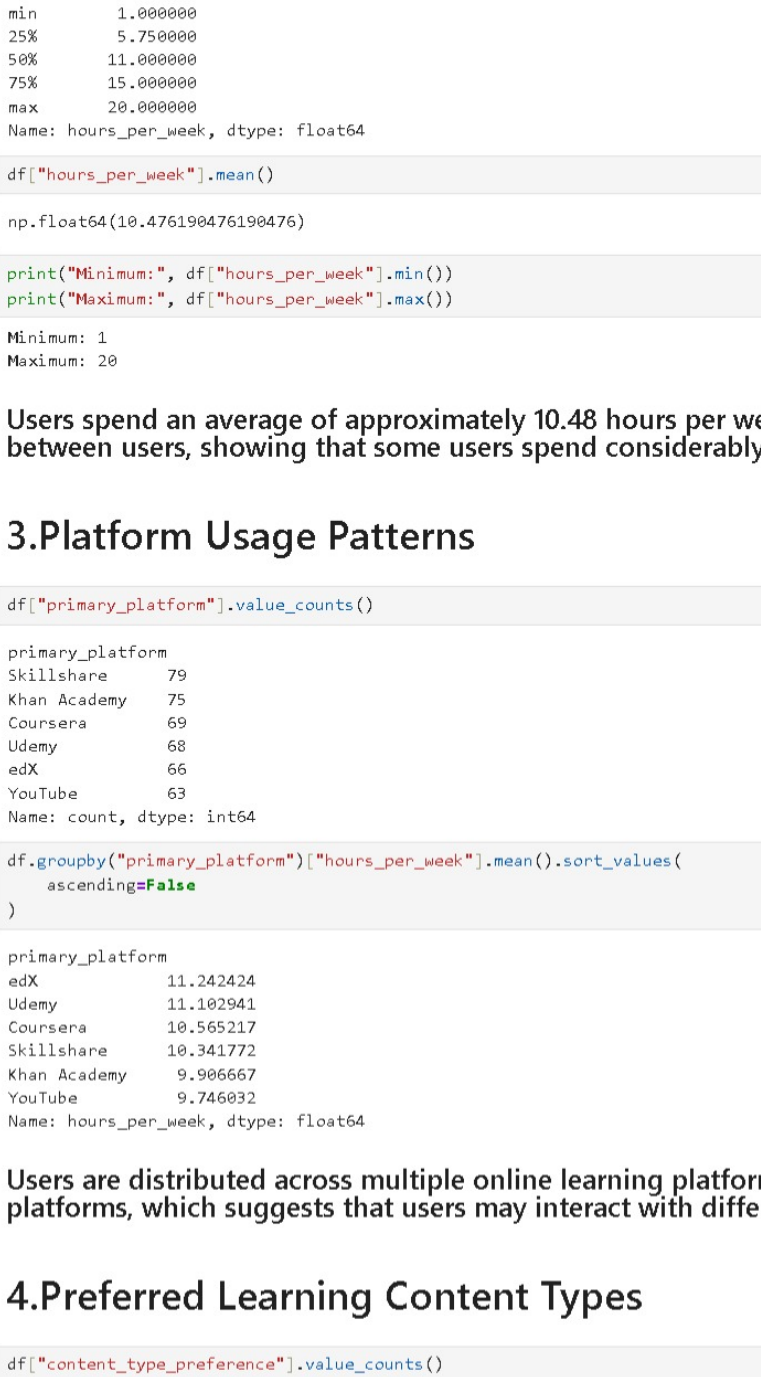
Completion Rate by Learning Purpose

```
[9]: purpose_completion = df.groupby(
        "learning_purpose"
    )["completion_rate_percent"].mean().sort_values(
        ascending=False)

purpose_completion.plot(kind="bar")

plt.title("Average Completion Rate by Learning Purpose")
plt.xlabel("Learning Purpose")
plt.ylabel("Average Completion Rate (%)")

plt.show()
```



Users learning for Skill Upgrade have the highest average completion rate. This suggests that users with a specific practical goal may be more likely to complete their learning.

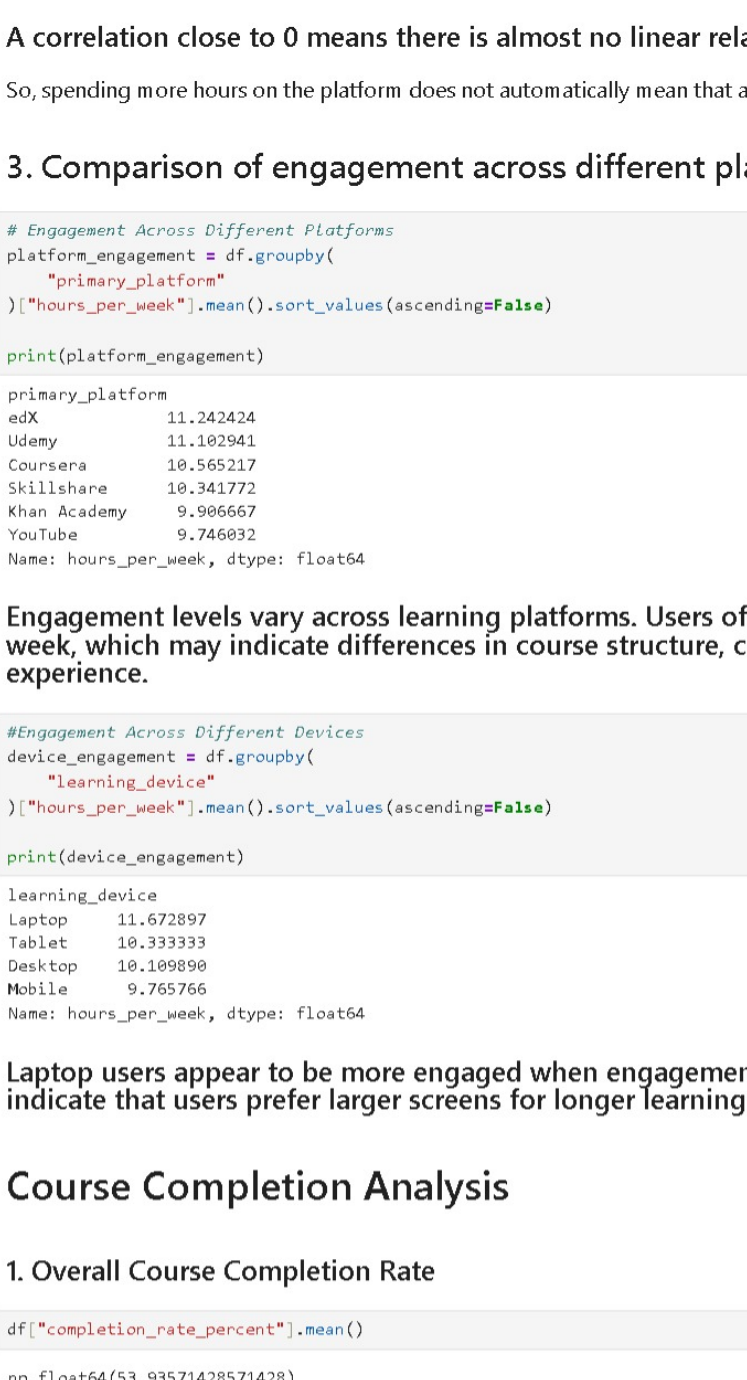
Satisfaction by Device

```
[10]: device_satisfaction = df.groupby(
        "learning_device"
    )["satisfaction_score"].mean().sort_values(
        ascending=False)

device_satisfaction.plot(kind="bar")

plt.title("Average Satisfaction by Learning Device")
plt.xlabel("Learning Device")
plt.ylabel("Average Satisfaction Score")

plt.show()
```



Mobile users have the highest average satisfaction among the device groups. This suggests that the mobile experience is relatively well received by users.

Platform Usage Pattern

```
[11]: platform_users = df["primary_platform"].value_counts()

print(platform_users)

primary_platform
Skillshare    79
Khan Academy  75
Coursera      69
Udemy         68
edX           66
YouTube       63
Name: count, dtype: int64
```

```
[12]: platform_users.plot(kind="bar")

plt.title("Number of Users by Learning Platform")
plt.xlabel("Learning Platform")
plt.ylabel("Number of Users")

plt.show()
```



The chart shows how the user base is distributed across different learning platforms. This can help the company understand which platforms are commonly used by its customers.

Business Insights

1. The 35-44 age group has the highest number of users, with 98 users. This shows that this age segment forms an important part of the platform's user base.

2. Users aged 25-34 spend the highest average time learning, at around 11.62 hours per week. This group appears to be highly active on the platform.

3. Users whose main purpose is Skill Upgrade have the highest average completion rate, around 59.21%. Goal-oriented users appear more likely to complete their learning.

4. Desktop users have the highest average completion rate at around 58.29%, while mobile users report the highest average satisfaction score of around 5.59.

5. Users spend an average of around 10.48 hours per week learning. However, learning hours have almost no relationship with completion rate.

6. Satisfaction and completion rate are also only very weakly related. This means a user can be satisfied with the platform without necessarily completing more courses.

7. User behavior differs across age groups, platforms, devices and learning purposes. Therefore, the company should avoid using the same engagement strategy for every user.

Business Recommendations

1- Improve Completion Support

The platform should introduce progress reminders, milestone notifications and smaller learning targets to encourage users to complete their courses.

2- Personalize Learning

Learning recommendations should be based on the user's learning purpose, age group, device and previous activity instead of giving every user the same content.

3- Improve Desktop Learning

Since desktop users show a relatively high completion rate, the company can study which desktop features support course completion and consider bringing similar features to other devices.

4- Focus on User Engagement

Instead of measuring engagement only through hours spent, the company should also track course progress, completed lessons, return frequency and interaction with learning content.

5- Monitor Satisfaction Separately

Satisfaction and completion should be treated as separate KPIs because the analysis does not show a strong relationship between them. Regular feedback surveys can help identify specific user experience problems.